

Quantum Mechanics Problem Sheet 3 Solutions

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1 Question 1

i)

$$\lambda = \frac{h}{p} \text{ and } p = mv$$

Substituting gives $\lambda = \frac{h}{mv}$

Given that $m_e = 9.109 \times 10^{-31}$, $h = 6.626 \times 10^{-34}$ and $v = \frac{2.998 \times 10^8}{137}$

$$\lambda = \frac{6.626 \times 10^{-34}}{9.109 \times 10^{-31} \times \frac{2.998 \times 10^8}{137}} = 3.32 \times 10^{-10} \text{ m}$$

ii)

Rearranging the first equation gives $v = \frac{h}{\lambda m}$ h is a constant, and given that the wavelengths are the same, we can find the ratio of their velocities (electron to neutron).

$$\frac{v_e}{v_n} = \frac{m_n}{m_e} = \frac{1.6749 \times 10^{-27}}{9.109 \times 10^{-31}} = 1839$$

The ratio of their kinetic energies is $\frac{\frac{1}{2}m_e v_e^2}{\frac{1}{2}m_n v_n^2}$

We know from above that $\frac{v_e}{v_n} = \frac{m_n}{m_e}$.

Substituting and cancelling the half gives $\frac{m_e}{m_n} \left(\frac{m_n}{m_e}\right)^2$

Simplifying to $\frac{m_n}{m_e} = 1839$ so the ratio of their velocities is the same as the ratio of their kinetic energies.

iii)

Using the given approximations

$$\Delta x \Delta p \approx \frac{1}{2} \hbar$$

$$\Delta E \Delta t \approx \frac{1}{2} \hbar$$

$$\Delta p \approx mc$$

$$\Delta E \approx mc^2$$

We find the mass by $m = \frac{91.2 \times 10^9 \times 1.602 \times 10^{-19}}{(2.998 \times 10^8)^2} = 1.63 \times 10^{-25} \text{ kg}$

$$\Delta x mc = \frac{1}{2} \hbar$$

$$\Delta x = \frac{\hbar}{2mc} = \frac{h}{4\pi mc} = \frac{6.626 \times 10^{-34}}{4\pi \times 2.998 \times 10^8 \times 1.63 \times 10^{-25}} = 1.08 \times 10^{-18} \text{ m}$$

$$\Delta t = \frac{\hbar}{2mc^2}, \text{ which is } \frac{\Delta x}{c} = \frac{1.08 \times 10^{-18}}{2.998 \times 10^8} = 3.60 \times 10^{-27}$$

iv)

The wave equation is $\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$

$$\psi(x, t) = A \sin(kx - \omega t)$$

First we differentiate $\psi(x, t)$ with respect to x and constant t .

$$\frac{\partial \psi}{\partial x} = Ak \cos(kx - \omega t)$$

$$\frac{\partial^2 \psi}{\partial x^2} = -Ak^2 \sin(kx - \omega t) = -k^2 \psi(x, t)$$

Now we differentiate $\psi(x, t)$ with respect to t and constant x .

$$\frac{\partial \psi}{\partial t} = -A\omega \cos(kx - \omega t)$$

$$\frac{\partial^2 \psi}{\partial t^2} = -A\omega^2 \sin(kx - \omega t) = -\omega^2 \psi(x, t)$$

Plugging into the wave equation, we get $-k^2 \psi(x, t) = -\frac{1}{c^2} \omega^2 \psi(x, t)$

$$\implies k^2 \psi(x, t) = \frac{1}{c^2} \omega^2 \psi(x, t) \implies k^2 c^2 = \omega^2 \implies kc = \omega \text{ (if } \psi \text{ is a solution of the wave equation)}$$

Given that sine is periodic with period 2π , $k = \frac{2\pi}{\lambda}$, and $\omega = 2\pi f$

$$c = f\lambda$$

Combining our last two equations gives $\omega = \frac{2\pi}{\lambda} c$, but we know $\frac{2\pi}{\lambda} = k$, so we have shown that $\omega = kc$ as required, so ψ is a solution of the wave equation.

v)

Electrons have a de-Broglie wavelength given by $\lambda = \frac{h}{\sqrt{2meV}}$

If the electrons are not measured, they behave as waves, forming minima and maxima (bright and dark spots) on the screen following the equation $n\lambda = d \sin \theta$.

If the electrons are measured, there is a 'wavefunction collapse' and they behave as particles, so there can be no interference, and the path of the electron can be deduced.

vi)

$$E = \frac{1}{2}mv^2 \text{ and } p = mv$$

$$\frac{p}{m} = v$$

$$E = \frac{1}{2} \frac{p^2}{m^2} m$$

$$E = \frac{p^2}{2m}$$

vii)

$$\lambda = \frac{ax}{D}$$

$$n\lambda = d \sin \theta$$

$$\lambda = s \sin 30 \implies \lambda = \frac{1}{2}s = 1.075 \times 10^{-10} \text{ m}$$

Now we calculate the energy and speed of the neutrons.

$$E = \frac{p^2}{2m} \text{ and } p = \frac{h}{\lambda}$$

$$\text{Combining gives } E = \frac{h^2}{2m\lambda^2} = \frac{(6.626 \times 10^{-34})^2}{2 \times 1.6749 \times 10^{-27} \times (1.075 \times 10^{-10})^2} = 1.13 \times 10^{-20} \text{ J}$$

$$\text{Converting to eV: } \frac{1.13 \times 10^{-20}}{1.602 \times 10^{-19}} = 0.0708 \text{ eV}$$

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 1.13 \times 10^{-20}}{1.6749 \times 10^{-27}}} = 3673 \text{ m/s}$$

viii)

$$hf = \phi + E_{k,\max} \implies f = \frac{\phi + E_{k,\max}}{h} = \frac{(4.3+5.1)}{6.626 \times 10^{-34}} \times 1.602 \times 10^{-19} = 2.27 \times 10^{15} \text{ Hz}$$

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8}{2.27 \times 10^{15}} = 132 \text{ nm}$$

$$E = \frac{h^2}{2m_e \lambda^2} = \frac{(6.626 \times 10^{-34})^2}{2 \times 9.109 \times 10^{-31} \times (132 \times 10^{-9})^2} = 1.38 \times 10^{-23} \text{ J} = 8.64 \times 10^{-5} \text{ eV}$$

ix)

This means that the electrons have a de-Broglie wavelength of 0.42nm.

$$E = \frac{h^2}{2\lambda^2 m_e} = \frac{(6.626 \times 10^{-34})^2}{2 \times 9.109 \times 10^{-31} \times (0.42 \times 10^{-9})^2} = 1.37 \times 10^{-18} \text{ J} = 8.53 \text{ eV}$$

x)

$$\Delta x \Delta p \geq \frac{\hbar}{2} \text{ and } \Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\text{We get } \Delta t \geq \frac{\hbar}{2\Delta E}$$

$$\Delta t \geq \frac{h}{4\pi\Delta E}$$

$$\Delta t \geq \frac{6.626 \times 10^{-34}}{4\pi \times 5 \times 10^6 \times 1.602 \times 10^{-19}}$$

$$\text{So } \Delta t \geq 6.58 \times 10^{-23} \text{ s}$$

$$\text{In a similar way, } \Delta x \geq \frac{h}{4\pi\Delta p}$$

$$\Delta p = \sqrt{2Em_e} = \sqrt{2 \times 5 \times 10^6 \times 1.602 \times 10^{-19} \times 9.109 \times 10^{-31}} = 1.21 \times 10^{-21}$$

$$\text{So } \Delta x \geq \frac{6.626 \times 10^{-34}}{4\pi \times 1.21 \times 10^{-21}} = 4.36 \times 10^{-14}$$

2 Question 2

i)

We explain why standing waves of the form $\psi(x, t) = A_n \sin(\frac{n\pi x}{a}) \cos(\omega t)$ are suitable for the particle in a box model.

The particle is confined between $x = 0$ and $x = a$, so by the Born interpretation, the modulus of the wavefunction squared must be 0 if x is not in the range 0 to a .

If $x = 0$, the wavefunction is also zero (because $\sin(0) = 0$)

If $x = a$, the wavefunction is $\sin(n\pi)$, which is equal to zero (so long as $n \in \mathbb{Z}$).

The cosine term allows for time dependency of the wavefunction.

A standing wave, rather than a travelling wave, is appropriate as the 0 probabilities at $x = 0$ and $x = a$ implies there are nodes at the boundaries.

This means that $a = n\frac{\lambda}{2}$, so there is an integer number of wavelengths inside the box (to allow for the nodes), so that $\lambda = \frac{2a}{n}$.

From the wavenumber $k = \frac{2\pi}{\lambda}$, and $\lambda = \frac{2a}{n}$, we can deduce that the sine term must be $\sin(\frac{n\pi x}{a})$.

ii)

The time-independent Schrödinger equation (with $V = 0$ as we are inside the box) is $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$.

We first evaluate $\frac{\partial \psi}{\partial x} = A_n \frac{n\pi}{a} \cos(\frac{n\pi x}{a}) \cos(\omega t)$

We then get $\frac{\partial^2 \psi}{\partial x^2} = -A_n \frac{n^2 \pi^2}{a^2} \sin(\frac{n\pi x}{a}) \cos(\omega t) = -\psi \frac{n^2 \pi^2}{a^2}$

So using Schrödinger's equation, $E\psi = \psi \frac{n^2\pi^2}{a^2} \frac{\hbar^2}{2m}$

Giving $E(n, a) = \frac{n^2\pi^2\hbar^2}{2ma^2}$

For a proton with $a = 0.47 \times 10^{-15}$, $E = \frac{n^2\pi^2(\frac{6.626 \times 10^{-34}}{2\pi})^2}{2 \times 1.6726 \times 10^{-27} \times (0.47 \times 10^{-15})^2}$

Using $n = 1$ gives the energy as 1.49×10^{-10} J.

To convert into MeV, we do $\frac{1.49 \times 10^{-10}}{1.602 \times 10^{-19}} \times 10^{-6} = 928$ MeV

Rest mass energy of a proton in eV is $\frac{(2.998 \times 10^8)^2 \times 1.6726 \times 10^{-27}}{1.602 \times 10^{-19}} \times 10^{-6} = 940$ MeV

iii)

The time dependent Schrödinger equation is $-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = i\hbar \frac{\partial \psi}{\partial t}$.

We choose $e^{-\frac{iEt}{\hbar}}$ instead of $\cos(\omega t)$ so that when we differentiate with respect to t, for the RHS of the above equation, we have a derivative that's a constant multiplied by the original function, which works for exponentials.

We can see this by $\frac{\partial}{\partial t}(e^{-\frac{iEt}{\hbar}}) = -\frac{iE}{\hbar}e^{-\frac{iEt}{\hbar}}$

We now determine A_n by considering that the sum of the probabilities of a particle being at any x position between 0 and a must be one.

$$\int_0^a |\psi|^2 dx = 1$$

We now use the standard results that $|z|^2 = z\bar{z}$, and the conjugate of e^{ix} is equal to e^{-ix} .

$$\int_0^a \psi \bar{\psi} dx = \int_0^a A_n \sin\left(\frac{n\pi x}{a}\right) e^{\frac{iEt}{\hbar}} \times A_n \sin\left(\frac{n\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} dx = 1$$

The integrand simplifies to $\int_0^a \sin\left(\frac{n\pi x}{a}\right)^2 dx = \frac{1}{A_n^2}$

We use the result that $\sin^2 x = \frac{1 - \cos 2x}{2}$

$$\int_0^a 1 - \cos\left(\frac{2n\pi x}{a}\right) dx = \frac{2}{A_n^2}$$

$$\left[x - \frac{a \sin\left(\frac{2n\pi x}{a}\right)}{2\pi n}\right]_0^a = \frac{2}{A_n^2}$$

The left hand side evaluates to a, giving $a = \frac{2}{A_n^2}$, so $A_n = \sqrt{\frac{2}{a}}$

iv)

Due to the symmetries of the box, and the fact that momentum is a vector, the average momentum going one way is equal to the average in the opposite direction, so they cancel out giving $\langle p \rangle = 0$

Earlier we showed $E = \frac{p^2}{2m}$, so that $p^2 = 2mE$

$$\langle p^2 \rangle = 2m\langle E \rangle$$

We have from above that $E = \frac{n^2\pi^2\hbar^2}{2ma^2}$

Giving $\langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}$ as required.

v)

$$\langle x \rangle = \int_0^a x A_n^2 \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) dx$$

Using the given integrals (which can be derived using integration by parts), we get $\frac{2}{a} \left[\frac{1}{4}x^2 - \frac{x \sin\left(\frac{2n\pi x}{a}\right)}{4\frac{n\pi}{a}} - \frac{\cos\left(\frac{2n\pi x}{a}\right)}{8\frac{n^2\pi^2}{a^2}} \right]_0^a$

We note that $\forall n \in \mathbb{Z}$: $\sin(2n\pi) = 0$ and $\cos(2n\pi) = 1$, so that the sine and cosine term cancels out.

Evaluating the integral between a and 0 gives $\langle x \rangle = \frac{2}{a}(\frac{1}{4}a^2) = \frac{a}{2}$ as required.

We now look for $\langle x^2 \rangle = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx$

Again using the given integrals, we get $\frac{2}{a} \left[\frac{1}{6}x^3 - \frac{x \cos(\frac{2n\pi x}{a})}{4(\frac{n\pi}{a})^2} - \frac{(2(\frac{n\pi x}{a})^2 - 1) \sin(\frac{2n\pi x}{a})}{8(\frac{n\pi}{a})^3} \right]_0^a$

We note that again the sine term cancels, but the cosine term does not.

$$\frac{2}{a} \left[\frac{1}{6}x^3 - \frac{x \cos(\frac{2n\pi x}{a})}{4(\frac{n\pi}{a})^2} \right]_0^a = \frac{2}{a} \left(\left(\frac{1}{6}a^3 - \frac{a}{4\frac{n^2\pi^2}{a^2}} \right) - (0 - 0) \right) = \frac{2}{a} \left(\frac{1}{6}a^3 - \frac{a^3}{4n^2\pi^2} \right)$$

This simplifies to $\frac{1}{3}a^2 - \frac{a^2}{2n^2\pi^2}$, which factorises to $\frac{1}{3}a^2(1 - \frac{3}{2n^2\pi^2})$, as required.

vi)

We use $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ and $\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$, with our earlier results.

$$\Delta x = \sqrt{\frac{1}{3}a^2(1 - \frac{3}{2n^2\pi^2}) - \frac{1}{4}a^2} = \sqrt{\frac{a^2}{12} - \frac{a^2}{2n^2\pi^2}} = a\sqrt{\frac{n^2\pi^2 - 6}{12n^2\pi^2}} = \frac{a}{2n\pi}\sqrt{\frac{n^2\pi^2}{3} - 2}$$

$$\Delta p = \sqrt{\frac{\hbar^2\pi^2n^2}{a^2}} = \frac{\hbar n\pi}{a}$$

So that $\Delta x \Delta p = \frac{a}{2n\pi} \frac{\hbar n\pi}{a} \sqrt{\frac{n^2\pi^2}{3} - 2}$, which simplifies to $\frac{1}{2}\hbar\sqrt{\frac{n^2\pi^2}{3} - 2}$ as required.

We now show it satisfies the uncertainty principle, so that $\Delta x \Delta p \geq \frac{1}{2}\hbar$

Therefore, we only have to consider whether $\sqrt{\frac{n^2\pi^2}{3} - 2}$ is greater than one. It is an increasing function of $n \forall n \in \mathbb{N}$, so we consider the minimum value of the function at $n = 1$.

We get $\sqrt{\frac{\pi^2}{3} - 2}$, which evaluates to 1.14 (2 d.p.), which is certainly greater than one for all n , so the particle always satisfies the Uncertainty principle.

3 Question 3

We use the formulas derived from the challenge, which can be found in the PDF after downloading the 2026 challenge from this link.

We have $\theta = -45^\circ$ and $\phi = 30^\circ$

Malus' Law (Classical)

$$P(\text{match}) = \cos^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi$$

$$P(\text{mismatch}) = 1 - \cos^2 \theta \cos^2 \phi - \sin^2 \theta \sin^2 \phi$$

$$\text{So } P(\text{mismatch}) = 1 - \cos^2(-45) \cos^2(30) - \sin^2(-45) \sin^2(30) = 0.50$$

Copenhagen Interpretation (Quantum)

$$P(\text{match}) = \cos^2(\phi - \theta)$$

$$P(\text{mismatch}) = \sin^2(\phi - \theta)$$

$$\text{So } P(\text{mismatch}) = \sin^2(30 - -45) = \sin^2(75) = \frac{2+\sqrt{3}}{4} \approx 0.93$$